

ApexMaths — Full Application Walk-through

Next.js on Vercel · Amazon Aurora PostgreSQL (Serverless v2, RDS Data API) · Cognito · Bedrock · Stripe

A guided tour: the product, the architecture, the question pipeline, the data layer, payments & IaC.

Live: <https://vercel-v0-and-aws-hackathon.vercel.app/>

THE PRODUCT — landing

ApexMaths: AI-powered UK 11+ maths prep, accessible to every family — not just those who can afford a tutor.

code: `app/page.tsx`

 ApexMaths

[How it works](#) [Progress tracking](#) [Features](#) [Pricing](#)

[Sign in](#) [Start free trial](#)

 AI tutoring, without the £30/hour price tag

Give your child a fair shot at the 11+ maths exam

Timed, exam-style practice with an AI tutor that explains how — never just the answer. Our adaptive **Skill builder** tailors every session to your child, quietly targeting their weakest topics over time — plus clear topic-by-topic progress for you. All for £19.99/month, after a 7-day free trial.

[Start your 7-day free trial](#) [See how it works](#)

£19.99
per month

7 days
free trial

6 topics
full curriculum coverage



£30+/hr

Private tutoring is expensive

Traditional 11+ tutors charge £30–£50 an hour, putting structured preparation out of reach for many families.

Hours

Tracking progress by hand is hard

Marking worksheets and spotting weak topics manually is slow and easy to get wrong — so children practise the wrong things.

£19.99

A fraction of the cost of a tutor

ApexMaths gives your child structured, exam-style preparation for less than the price of a single tutoring hour — making good preparation

THE PRODUCT — what it offers

Four practice modes, an AI tutor, per-question review, and a self-explaining parent dashboard.

code: `app/(marketing) · components/`

 ApexMaths

[How it works](#)[Progress tracking](#)[Features](#)[Pricing](#)

[Sign in](#)[Start free trial](#)

FEATURES

Everything a focused 11+ plan needs

Designed with parents for authority and children for friendliness — all in one calm, distraction-free app.



Adaptive Skill builder
Builds a unique session for each child from their own results — weighting more questions onto their weakest topics and pitching the difficulty just right, so they improve where it matters, session after session.



AI review report, per child
A written progress report generated on demand for each child from their data in that moment — strengths, the topics to focus on, and concrete next steps for the week.



“Show me how” AI tutor
Stuck on a question? Your child gets a clear, step-by-step explanation of the method — and it never reveals the answer.



Improves over time
Every answer sharpens the picture, so each adaptive session targets your child’s current weak spots — steady, compounding progress instead of random practice.



Topic-level progress
See mastery scores across all six curriculum topics, with each marked strong, developing, or needs focus.



Private & GDPR-first
Children’s names are never sent to the AI. Export or delete your data at any time, with one click.

Full coverage of the 11+ maths curriculum

 Number

 Fractions, Decimals & %

 Ratio & Proportion

 Algebra

 Geometry

 Data Handling

THE PRODUCT — sign-up & verification

Amazon Cognito email verification. No Cognito client secret; ID tokens verified against the pool JWKS.

code: `app/(auth)/sign-up` · `lib/auth/cognito.ts`

The screenshot shows a web browser window with the URL `vercel-v0-and-aws-hackathon.vercel.app/sign-up`. The browser's address bar and tabs are visible at the top. The page features the ApexMaths logo in the top left corner. The main content area is a light blue gradient with a central white card titled "Create your parent account". Below the title, a subtitle reads: "One account manages all of your children. Start with a free trial — no card required to explore." A green message box states: "We sent a 6-digit code to `mlrkovic.uk@gmail.com`. Enter it below to verify your account." Below this, a "Verification code" label is followed by a text input field containing "123456". A blue "Verify email" button is positioned below the input field. A link "Didn't get it? Resend code" is located below the button. At the bottom of the card, a link "Already have an account? Sign in" is present. A footer note at the very bottom of the page states: "Protected by Amazon Cognito. We never store your password."

vercel-v0-and-aws-hackathon.vercel.app/sign-up

Welcome from... Contact info First Alternative BEXLEY 11 PL... Free Materials... AWS Marketpl... AWS Builder C... Sign in | eAdmi... Sparx Maths St... Primary Hacka...

All Bookmarks

ApexMaths

Create your parent account

One account manages all of your children. Start with a free trial — no card required to explore.

We sent a 6-digit code to `mlrkovic.uk@gmail.com`. Enter it below to verify your account.

Verification code

123456

Verify email

[Didn't get it? Resend code](#)

Already have an account? [Sign in](#)

Protected by Amazon Cognito. We never store your password.

THE PRODUCT — verification confirmed

Email verified — the account is now active and can sign in.

code: `app/(auth)/sign-in · lib/auth/session.ts`

← → ↻ 🏠

vercel-v0-and-aws-hackathon.vercel.app/sign-in?verified=1

Screencast recorded
Click here to view the video.

🔑 🔍 ☆ 🏠 🧑

📁 All Bookmarks

📁

Welcome from...

👤 Contact info

🔗 First Alternative

📄 BEXLEY 11 PL...

📄 Free materials...

📄 AWS Marketpl...

📄 AWS Builder C...

📄 Sign in | eAdmi...

📄 Sparx Maths St...

📄 Primary Hacka...

»

📁 All Bookmarks

ApexMaths

🔑

Welcome back

Sign in to view your children's progress and start a practice session.

🔑

Your email is verified. Please sign in.

Email address

Password

[Forgot password?](#)

Sign in

New to ApexMaths? [Create an account](#)

🔑

Protected by Amazon Cognito. We never store your password.

THE PRODUCT — identity is real (Cognito console)

The verified user in the Cognito user pool (status CONFIRMED) — identity is managed by AWS, not in our DB.

code: `infra/lib/apexmaths-stack.ts` (User Pool)

The screenshot shows the Amazon Cognito console interface. The left sidebar contains navigation links for Amazon Cognito, User pools, apexmaths-users, and Users. The main content area is titled 'Users (6)' and includes a search bar and a table of users. The table has columns for User name, Email address, Email verified, Confirmation status, and Status. Below the table is the 'Import users (0)' section, which includes a search bar and a table for import jobs. The footer of the console shows the AWS logo, CloudShell, Feedback, Console Mobile App, and copyright information.

Users (6) Info

View, edit, and create users in your user pool. Users that are enabled and confirmed can sign in to your user pool.

Property: **User name**

	User name	Email address	Email verified	Confirmation status	Status
☐	b662f2a4-4071-7044-4aa4-cc31341f3...	lunim612@gmail.com	Yes	Confirmed	Enabled
☐	d6e2a284-8091-7083-2deb-cc36ba642...	demo_user1@apexmaths.com	No	Confirmed	Enabled
☐	9622e294-5091-7008-7cb3-5bb02152...	demo_admin@apexmaths.com	No	Confirmed	Enabled
☐	d68272f4-d061-70a6-0186-c5ee1aa77...	uros1311@gmail.com	Yes	Confirmed	Enabled
☐	26f23264-0051-708d-7c99-bb2e8a6f6...	demo_user@apexmaths.com	No	Confirmed	Enabled
☐	a642a284-6071-708c-e4f9-0d7f3ba38...	mirkovic.uk@gmail.com	Yes	Confirmed	Enabled

Import users (0) Info

View and create user CSV import jobs. Amazon Cognito can import users into this user pool from a specially-formatted CSV file. You can't import user passwords.

Job name	Status	Imported users	Skipped users	Failed users	CloudWatch logs	Created time
No user import jobs found						

[Create import job](#)

THE PRODUCT — onboarding gate

Guardian + age attestation before any app access. Authorization is layered and enforced before a page renders.

code: [app/\(auth\)/onboarding/page.tsx](#) · [lib/auth/guard.ts](#)

The screenshot shows a web browser window with the URL `vercel-v0-and-aws-hackathon.vercel.app/onboarding`. The browser's address bar and tabs are visible at the top. The main content area features the ApexMaths logo in the top left corner. Centered on the page is a white card with a light gray border. Inside the card, the heading 'Just one quick step' is followed by a paragraph: 'ApexMaths is managed by adults on behalf of children. Please confirm the following before you continue.' Below this text are two checkboxes, both of which are checked. The first checkbox is labeled 'I am the parent or legal guardian of the child or children who will use this account, and I consent to them practising on this platform.' The second checkbox is labeled 'I am 18 years or older and I will manage all account settings, billing and personal data on behalf of my child.' At the bottom of the card is a blue button with the text 'Continue to dashboard'. Below the card, there is a small green shield icon followed by the text 'Protected by Amazon Cognito. We never store your password.'

vercel-v0-and-aws-hackathon.vercel.app/onboarding

Screencast recorded
Click here to view the video.

Welcome from... Contact info First Alternative BEXLEY 11 PL... Free materials... AWS Marketplace AWS Builder C... Sign in | Admin... Sparx Maths St... Primary Hacka...

All Bookmarks

ApexMaths

Just one quick step

ApexMaths is managed by adults on behalf of children.
Please confirm the following before you continue.

- ☒ I am the parent or legal guardian of the child or children who will use this account, and I consent to them practising on this platform.
- ☒ I am 18 years or older and I will manage all account settings, billing and personal data on behalf of my child.

Continue to dashboard

Protected by Amazon Cognito. We never store your password.

THE PRODUCT — practice modes

Warm-up, single-topic, full mock, and the adaptive Skill Builder (today's mix tuned to the child's weak topics).

code: `app/(app)/practice/new` · `lib/practice/adaptive-selection.ts`

ApexMaths

DashboardAdminUR

[Back to dashboard](#)

Start a session

Pick who's practising and the type of session. Everything is timed and marked automatically.

Who's practising?

NI
Nina
Year 6

AM
Amara
Year 5 ✓

LU
Lui
Year 4

Choose a session

Warm-up

10 questions across mixed topics in 10 minutes.

🕒 10 ⌚ 10 min

Practice a topic

5 questions on a single topic in 10 minutes.

🕒 5 ⌚ 10 min

Full mock

30 questions across mixed topics in 50 minutes.

🕒 30 ⌚ 50 min

Skill builder

15 questions tuned to your child's level across all topics in 20 minutes.

🕒 15 ⌚ 20 min

The timer starts as soon as the session begins and runs on our server, so closing the tab won't pause it.

CancelStart session

THE PRODUCT — answering a question

A question — the correct index is NEVER sent to the client; grading is server-authoritative.

code: `app/(app)/practice/[sessionId]/page.tsx` · `app/(app)/practice/actions.ts`

ApexMaths

Dashboard Admin UR

Amara · Warm-up

9:19 Cancel

1/10

QUESTION 1 OF 10

Show me how (4 left)

A baker makes 12 cupcakes every hour. If we use the letter h to stand for the number of hours, which expression tells us the total number of cupcakes made?

A $h + 12$

B $12 \times h$ ✓

C $12 + h + 12$ ✗

D $h + 12$

E $12 - h$

Not quite. The right answer is B. You'll get a full step-by-step explanation in your review at the end.

< Previous

Next >

THE PRODUCT — AI tutor

'Show me how' streams a step-by-step method without revealing the answer; ask again for a different method.

code: [app/api/practice/help/route.ts](#) · [lib/ai/model.ts](#)

The screenshot displays the ApexMaths AI tutor interface. At the top, the ApexMaths logo is on the left, and navigation links for 'Dashboard', 'Admin', and 'UR' are on the right. The main content area shows a 'Warm-up' session with a timer at 9:42 and a progress bar at 0/10. A question titled 'A baker makes 12 cupcakes every single hour. How many cupcakes does the baker make in h hours?' is displayed. Below the question are five multiple-choice options: A $h + 12$, B $12 \times h$, C $12 + h$, D $h \times 12$, and E $12 - h$. A 'Show me how' button is visible. A modal window titled 'Show me how' is open, providing a step-by-step walkthrough. The modal text includes: 'A step-by-step walkthrough — try to follow each step, then have another go. 4 hints left this session.' followed by five numbered steps explaining the logic of multiplication. At the bottom of the modal, it asks 'Which option matches what you wrote? Have a go!' and provides a 'Try a different way' button with 4 hints left. The background question is partially obscured by the modal.

ApexMaths

Dashboard Admin UR

Amara · Warm-up 9:42 × Cancel 0/10

QUESTION 1 OF 10 Show me how (4 left)

A baker makes 12 cupcakes every single hour. How many cupcakes does the baker make in h hours?

A $h + 12$
B $12 \times h$
C $12 + h$
D $h \times 12$
E $12 - h$

Show me how

A step-by-step walkthrough — try to follow each step, then have another go. 4 hints left this session.

1. Think about what is happening: the baker makes 12 cupcakes every single hour.
2. Ask yourself: If the baker works for 2 hours, you would add 12 twice. If they work for 3 hours, you would add 12 three times. That is the same as multiplying.
3. So the rule is: cupcakes per hour $\times$ number of hours = total cupcakes.
4. We know cupcakes per hour is 12, and the letter h stands for the number of hours.
5. Write that as an expression using 12 and h with a multiplication sign.

Which option matches what you wrote? Have a go!

Still stuck? Try the same idea explained a different way. Try a different way (4 left)

< Previous Next >

THE PRODUCT — instant score

The score is instant and server-authoritative (first answer wins; timer enforced on the server).

code: `app/(app)/practice/[sessionId]/result/page.tsx` · `lib/db/sessions.ts`

The screenshot shows the ApexMaths application interface. At the top, there is a navigation bar with the ApexMaths logo, a 'Dashboard' link, an 'Admin' link, and a user profile icon labeled 'UR'. The main content area features a central card titled 'Room to grow' with a trophy icon, indicating a warm-up session for 'Amara'. The score is displayed as '0 / 10' with '0% correct'. Below this, a section titled 'How each topic went' shows a grid of topic performance boxes: Algebra (0/2), Geometry (0/1), Ratio & Proportion (0/1), Data Handling (0/1), Fractions, Decimals & Percentages (0/1), and Number (0/4). The 'Review answers' section lists two questions: 'A baker makes 12 cupcakes every hour. If we use the letter h to stand for the number of hours...' and 'How many lines of symmetry does a square have?'. The interface is clean with a light blue background and white cards.

THE PRODUCT — per-question AI review

Every wrong answer gets a plain-language AI explanation and a concrete next step.

code: [lib/ai/review.ts](#) · [lib/db/reviews.ts](#)

The screenshot displays the ApexMaths application interface. At the top, there is a navigation bar with the ApexMaths logo, a 'Dashboard' link, an 'Admin' link, and a user profile icon labeled 'UR'. The main content area is divided into two sections, each containing a math question with multiple-choice options, an AI-generated explanation, and a 'Next step' recommendation.

Question 1: A baker makes 12 cupcakes every hour. If we use the letter h to stand for the number of hours, which expression tells us the total number of cupcakes made?

Options:

- A $h + 12$
- B $12 \times h$ (Correct)
- C $12 + h + 12$ (Incorrect)
- D $h \div 12$
- E $12 - h$

Explanation: When the same amount is repeated over and over (12 cupcakes every hour), we use multiplication to find the total. So if the baker works for h hours, we multiply 12 by h to get $12 \times h$. Adding $12 + h + 12$ mixes up adding and multiplying — it doesn't show the cupcakes growing by 12 for each hour.

Next step: Try making up your own "rate" problems — like "a car travels 30 miles every hour" — and practise writing the expression using $\times$ instead of $+$. This will help the difference between adding and multiplying really click!

Question 2: How many lines of symmetry does a square have?

Options:

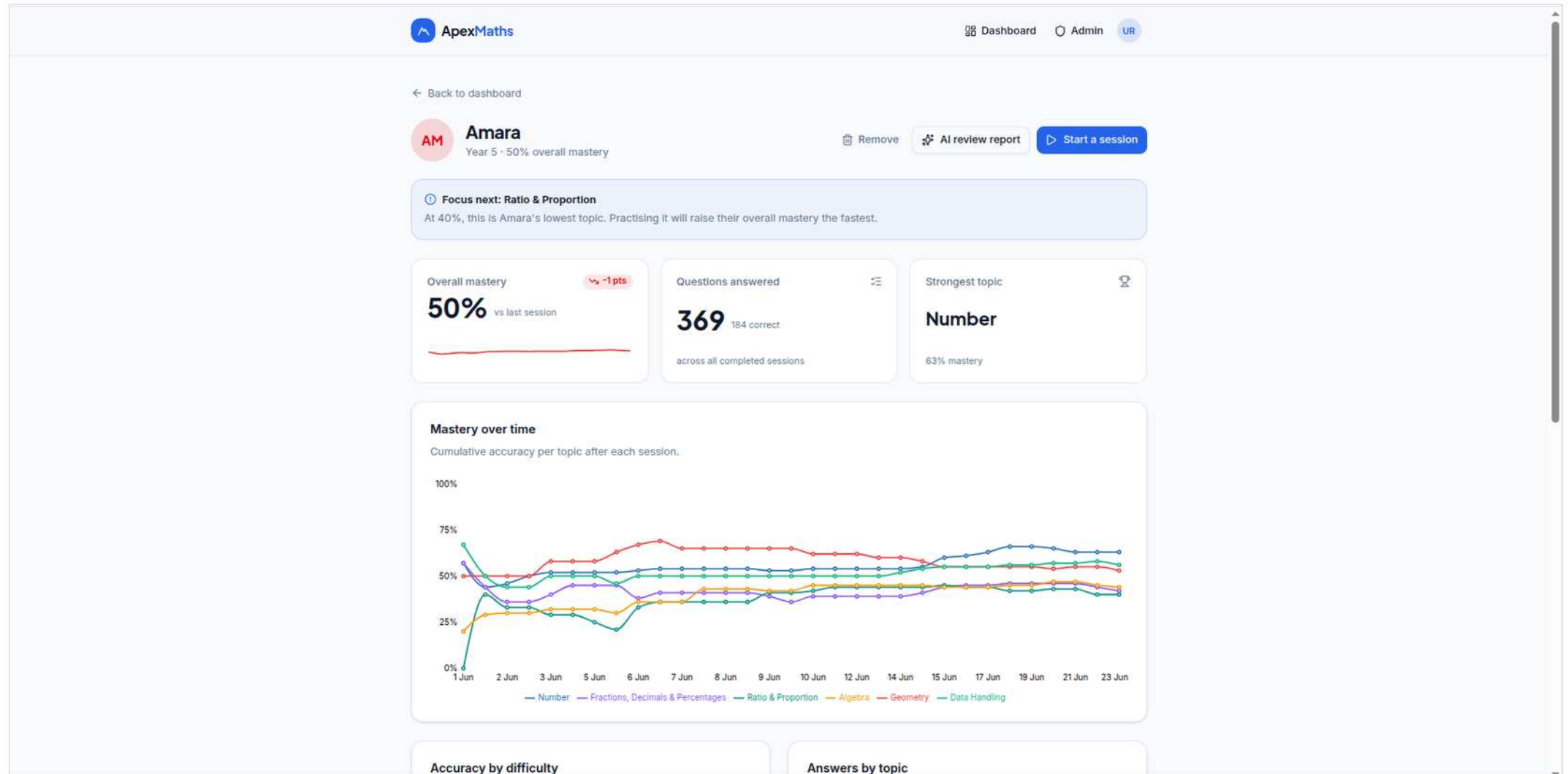
- A 1
- B 2
- C 3
- D 4 (Correct)
- E 8

Explanation: No worries — this one was left blank, but let's look at how to solve it! A line of symmetry is a line that splits a shape into two identical mirror-image halves. A square has 4 lines of symmetry.

THE PRODUCT — child analytics

Mastery over time and accuracy by difficulty — computed live in Postgres with window functions, no ETL.

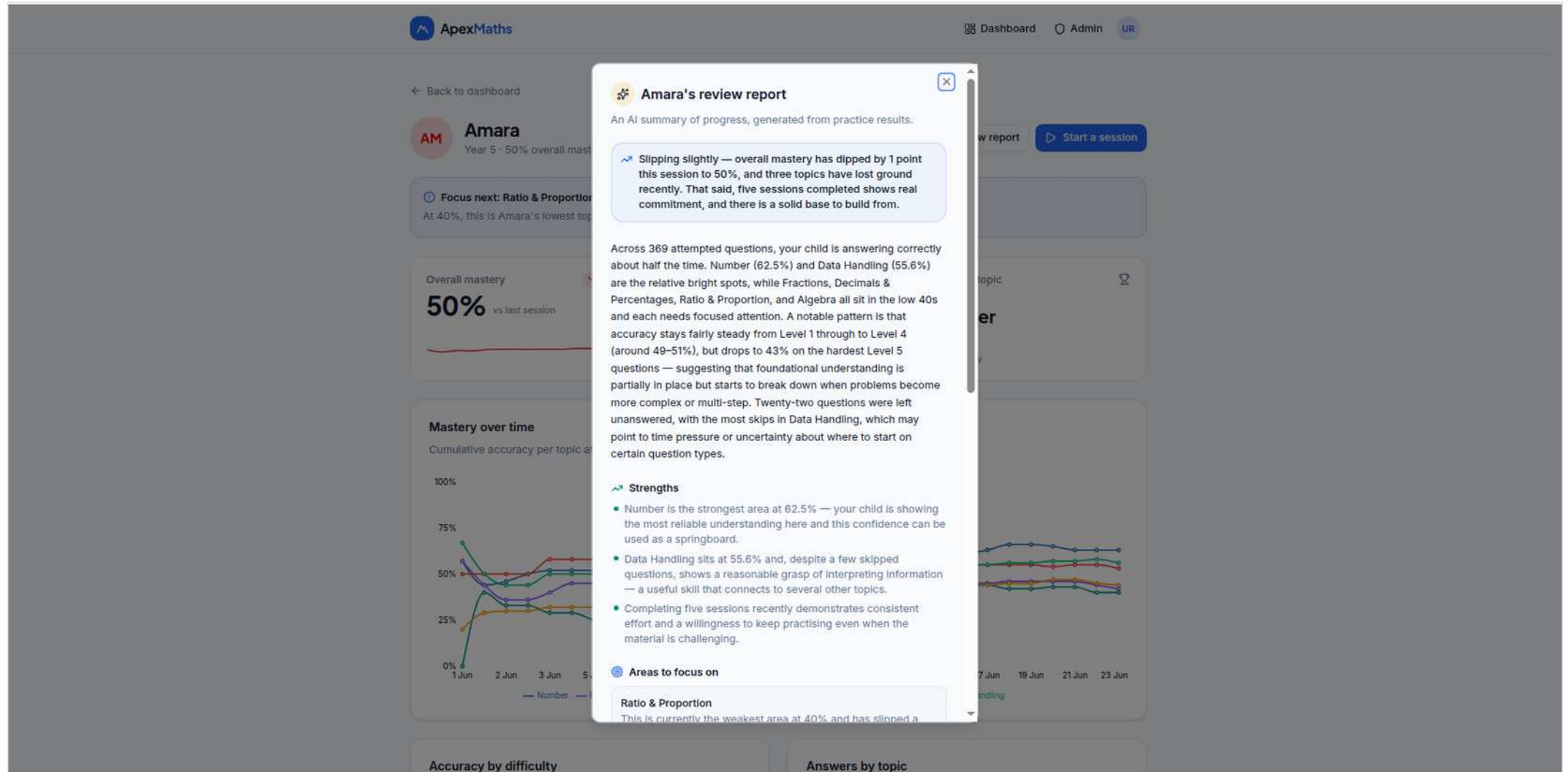
code: `lib/db/analytics.ts` · `app/(app)/children/[childId]/page.tsx`



THE PRODUCT — AI dashboard summary

AI reads the whole dashboard and writes a plain-English summary — turning data into guidance for a busy parent.

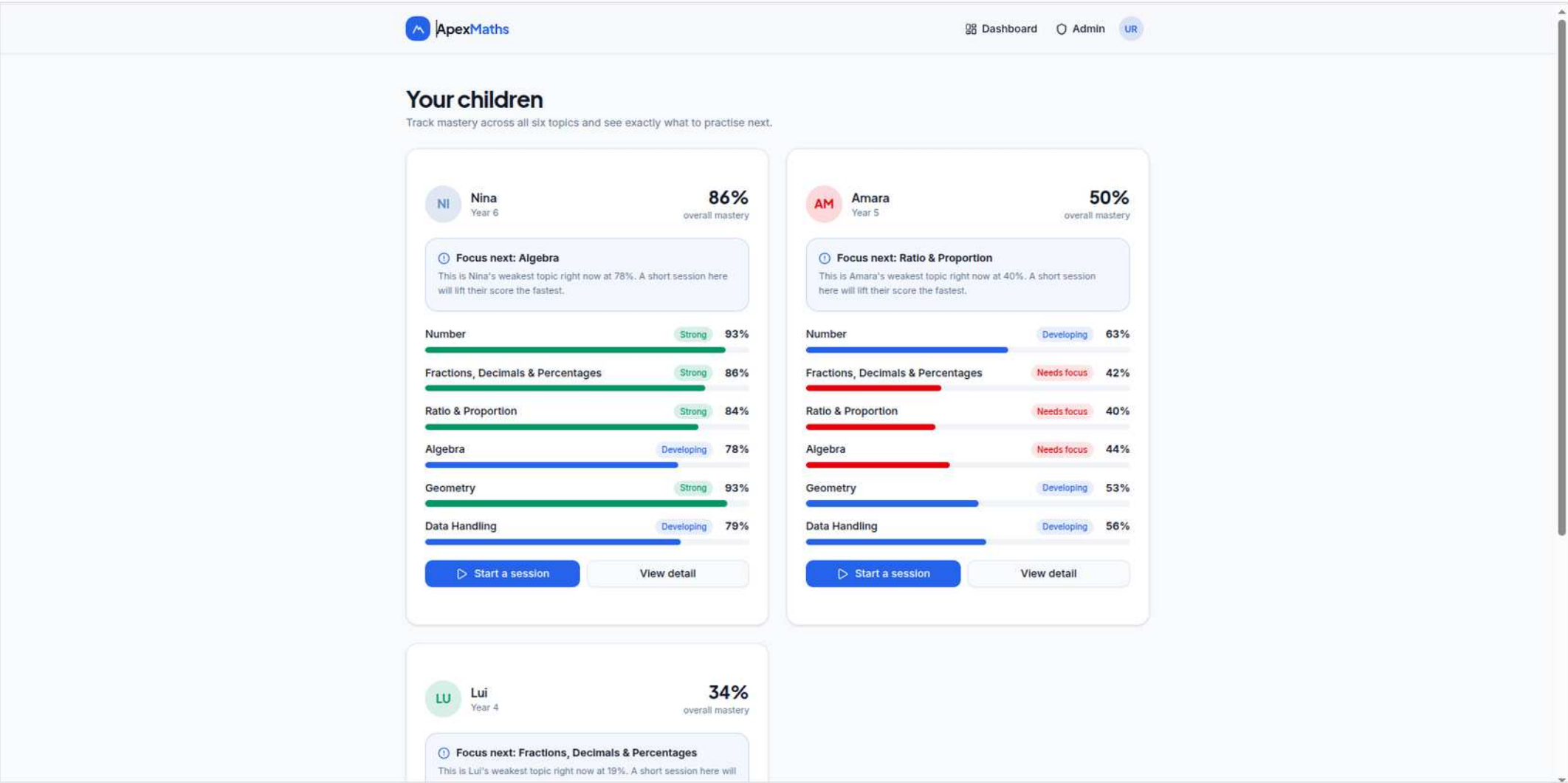
code: [lib/ai/report.ts](#) · [app/api/children/report/route.ts](#)



THE PRODUCT — parent view

Up to three child profiles, each with independent progress.

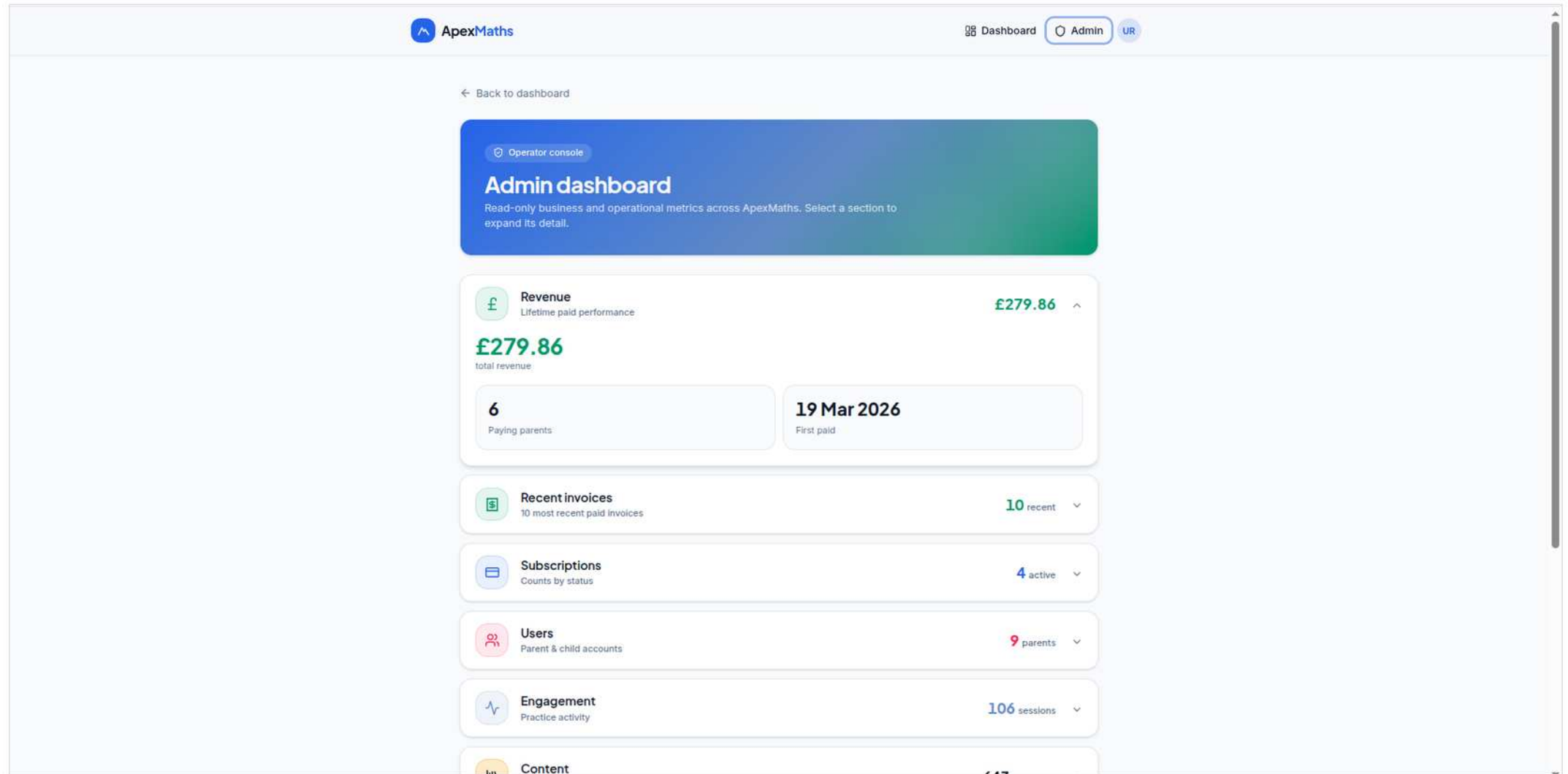
```
code: app/(app)/dashboard/page.tsx · lib/db/children.ts
```



THE PRODUCT — operator/admin console

Revenue, subscriptions, engagement, at-risk learners. Admin is fail-closed (non-admins get a 404).

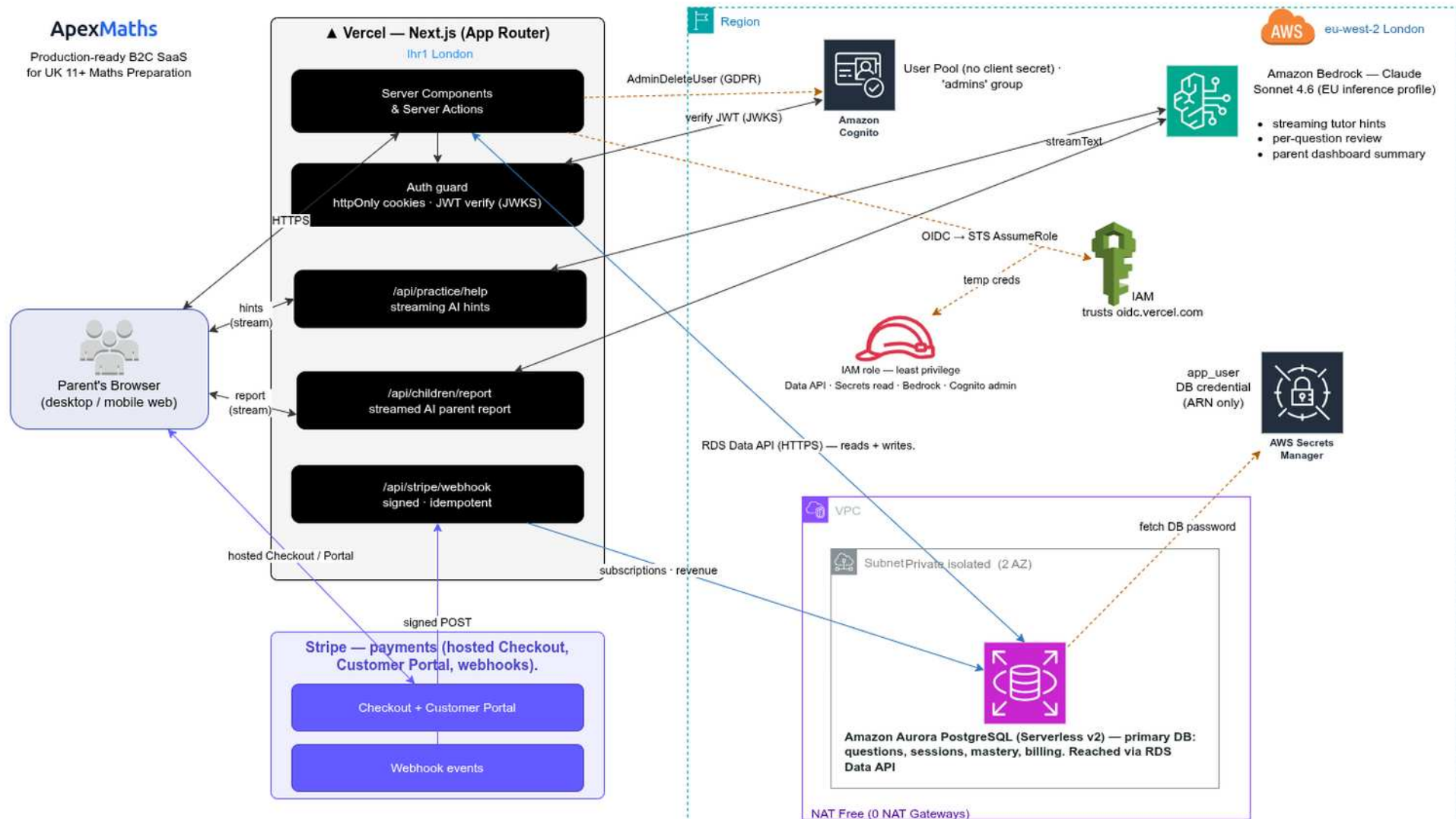
code: `app/(app)/admin/page.tsx` · `lib/db/admin-metrics.ts` · `lib/db/at-risk.ts`



ARCHITECTURE — runtime

Vercel is the entire backend; AWS holds data (Aurora), identity (Cognito), and AI (Bedrock).

code: —



Built for: H0: Hack the Zero Stack with Vercel v0 & AWS Databases by Aurora75 (devpost)

ARCHITECTURE — Vercel taken further (1/2)

All four SDKs run server-side. Vercel reaches AWS via OIDC federation — short-lived STS creds, no static keys.

code: `lib/aws/credentials.ts`

Aurora75's proj...

Hobby

vercel-v0-and-aws-hackathon

Project Settings

Security

Find...

F

<

Settings

General

Build and Deployment

Environments

Git

Deployment Protection

Passport

Functions

Cron Jobs

Microfrontends

Project Members

Drains

Security

Networking

Advanced

Learn more about [Deployment Protection](#)

Save

Secure Backend Access with OIDC Federation

Authenticate with your cloud providers using OpenID Connect JSON Web Tokens.

Issuer Mode

Team

Recommended

<https://oidc.vercel.com/aurora75-s-projects>

Global

<https://oidc.vercel.com>

Claim	Kind	Value
iss	Issuer	https://oidc.vercel.com/aurora75-s-projects
aud	Audience	https://vercel.com/aurora75-s-projects
sub	Subject	owner:aurora75-s-projects:project:vercel-v0-and-aws-hackathon:environment:production
scope	Scope	owner:aurora75-s-projects:project:vercel-v0-and-aws-hackathon:environment:production
iat	Issued At	1782654969
nbf	Not Before	1782654969
exp	Expiration	1782658569

Learn more about [Secure Backend Access with OIDC Federation](#)

Save

Deployment Retention Policy

Define a retention period for all Deployments for this Project. Deployments will be deleted after the defined time.

Canceled Deployments

Aurora75

...

🔔

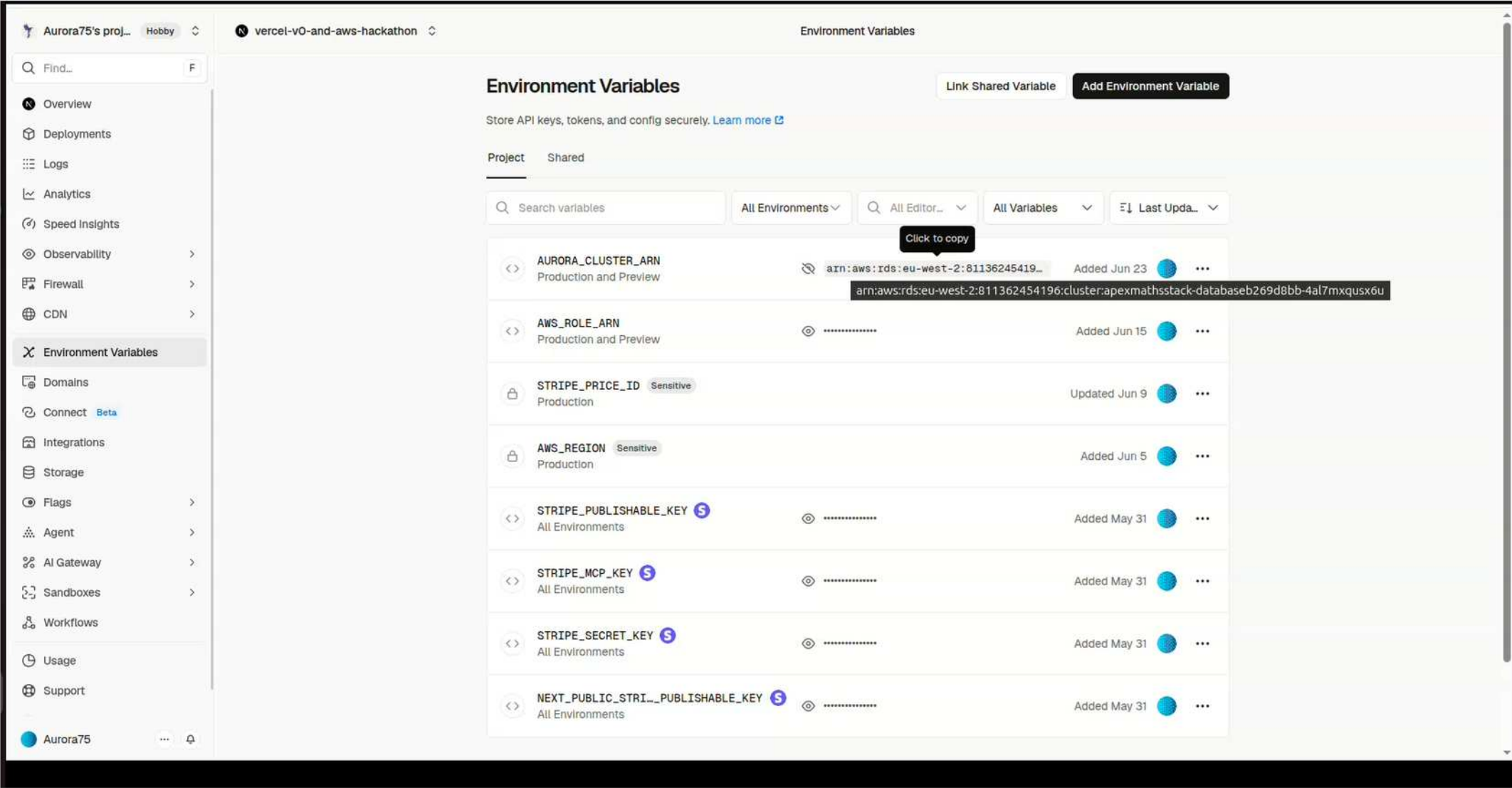
ApexMaths · Full Application Walk-through

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ARCHITECTURE — Vercel taken further (2/2)

The app targets Aurora by ARN over the RDS Data API — no connection pool, no VPC ingress, no password in code.

code: `lib/aws/rds-data.ts`



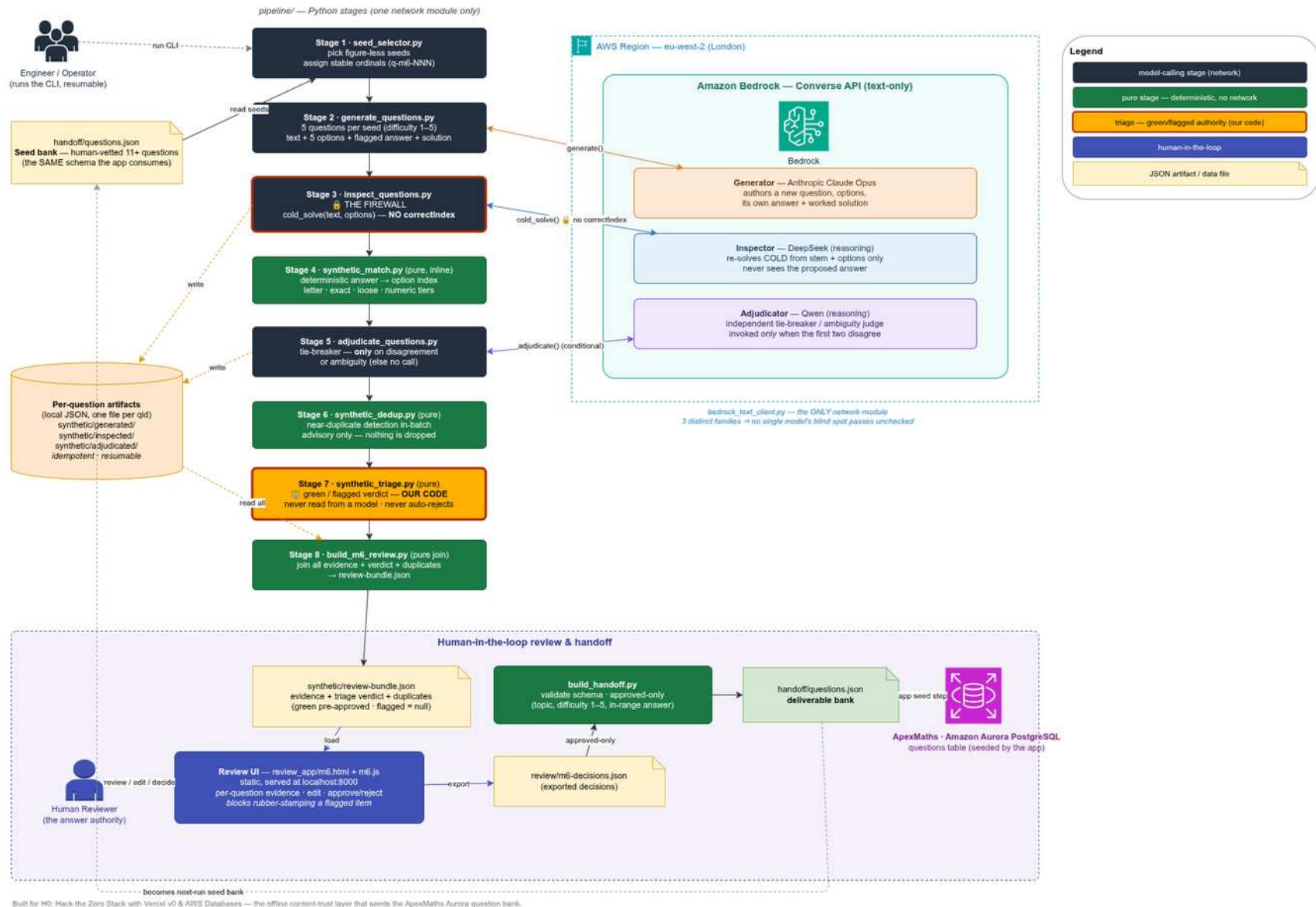
QUESTION PIPELINE — why & how

I needed trustworthy questions, so I built an offline pipeline: three independent model families propose; deterministic code judges; a human disposes.

code: [QuestionGenerationPipeline/pipeline/](#)

ApexMaths — Question Generation Pipeline

Offline, adversarial, human-gated factory that fills the production question bank. Three independent model families propose; deterministic code judges; a human disposes.



QUESTION PIPELINE — human review UI

The review app shows all three models' evidence per question; the human is the final answer authority.

code: `pipeline/build_m6_review.py` · `pipeline/review_app/m6.html`

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localhost:8000/pipeline/review_app/m6.html

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Welcome from...

Contact info

First Alternative

BEXLEY 11 PL...

Free Materials...

AWS Marketpl...

AWS Builder C...

Sign In | eAdmi...

Sparx Maths St...

Primary Hacka...

»

All Bookmarks

M6 Synthetic Question Review q-m6-010 · 2/216 shown · 124/620 decided

← Prev

#

Next →

☒ only needs-review

Export decisions

🚩

✎

✅

q-m6-010 · seed q-m1-003 · number · difficulty 4

A scientist writes down a number where the digit 4 appears exactly three times. The number is 4,304,042. What is the sum of all the values represented by the digit 4 in this number?

A 4,004,040 ✓

B 4,004,004

C 4,400,040

D 4,004,042

E 4,040,004

Generator solution

Step 1: Identify the positions of the digit 4 in 4,304,042.

The number is: 4,304,042.
Positions (from left to right):

- 4 is in the millions place (value = 4,000,000)
- 3 is in the hundred-thousands place
- 0 is in the ten-thousands place
- 4 is in the thousands place (value = 4,000)
- 0 is in the hundreds place
- 4 is in the tens place (value = 40)
- 2 is in the units place

Step 2: Add up all the values represented by the digit 4.
4,000,000 + 4,000 + 40 = 4,004,040

Step 3: The answer is 4,004,040.

Inspector (cold solve)

answer: A
matchedIndex: 0
answersAgree: true
difficulty: 2
exactlyOneCorrect: true

The number is 4,304,042. Write it without commas: 4304042. Identify the positions of each digit 4 from left to right:
1st digit from left: 4 in the millions place → value = 4 × 1,000,000 = 4,000,000.
3rd digit from left: 4 in the ten-thousands place → value = 4 × 10,000 = 40,000.
5th digit from left: 4 in the tens place → value = 4 × 10 = 40.
Sum = 4,000,000 + 40,000 + 40 = 4,040,040.
Check options: A) 4,004,040, B) 4,004,004, C) 4,400,040, D) 4,004,042, E) 4,040,004.
None match 4,040,040 exactly. Re-examine number: 4,304,042 → digits: 4 (millions), 3 (hundred-thousands), 0 (ten-thousands), 4 (thousands), 0 (hundreds), 4 (tens), 2 (ones). Wait: digit positions in full: millions: 4 (value 4,000,000), hundred-thousands: 3, ten-thousands: 0, thousands: 4 (value 4,000), hundreds: 0, tens: 4 (value 40), ones: 2. So three 4s: at millions (4,000,000), thousands (4,000), tens (40). Sum = 4,000,000 + 4,000 + 40 = 4,004,040. That matches option A exactly. So answer is A.

Adjudicator

Not adjudicated (solvers agreed, unambiguous).

Triage

🚩

• difficulty_gap

Duplicates

None.

DECISION

☒ Approve

☐ Reject

☒ I reviewed this flagged item and confirm it is correct.

▼ Adjust question (stem / options / answer / topic / difficulty)

Stem

A scientist writes down a number where the digit 4 appears exactly three times. The number is 4,304,042.

A 4,004,040

B 4,004,004

C 4,400,040

D 4,004,042

E 4,040,004

Correct option

A

Topic

number

Difficulty

4

NOTE

optional

QUESTION PIPELINE — code-authored triage

Triage (our code, never a model) flags disagreement/ambiguity; the human rejects or edits. Approved items seed Aurora.

code: [pipeline/synthetic_triage.py](#) · [pipeline/build_handoff.py](#)

← → ↺ 🏠

localhost:8000/pipeline/review_app/m6.html

🔍 ☆ 📁 🧑

🧩 | 🏠 Welcome from... 🕒 Contact info 📄 First Alternative 📄 BEXLEY 11 PL... 📄 Free Materials... 📄 AWS Marketpl... 📄 AWS Builder C... 📄 Sign In | eAdmi... 📄 Sparx Maths St... 📄 Primary Hacka... >> | 📁 All Bookmarks

M6 Synthetic Question Review q-m6-050 · 16/216 shown · 124/620 decided

⏪ Prev # Next ⏩ ☒ only needs review Export decisions

🚩 flagged reject

q-m6-050 · seed q-m1-022 · fractions_decimals_percentages · difficulty 5

A shop reduces the price of a jacket from £80 to £62. They then offer an additional 20% off the already-reduced price during a flash sale. What is the total percentage discount from the original price of £80?

A 22.5%

B 38.75% ✓

C 42.5%

D 36.25%

E 40%

Generator solution

Step 1: Find the first reduction. The price dropped from £80 to £62, so the first reduction is $£80 - £62 = £18$.

Step 2: Find the additional 20% off the reduced price. 20% of £62 = $0.20 \times £62 = £12.40$.

Step 3: Find the final sale price. $£62 - £12.40 = £49.60$.

Step 4: Find the total discount from the original price. $£80 - £49.60 = £30.40$.

Step 5: Express this as a percentage of the original price. $(£30.40 / £80) \times 100 = 38\%$.

Wait — let me recalculate more carefully: $30.40 / 80 = 0.38$, and $0.38 \times 100 = 38\%$.

Hmm, 38% is not among the options. Let me recheck: 20% of £62 = £12.40. Total discount = $£18 + £12.40 = £30.40$. As a percentage of £80: $(30.40/80) \times 100 = 38\%$.

Let me adjust: The shop reduces from £80 to £62, then takes an additional 25% off the reduced price. 25% of £62 = £15.50. Final price = $£62 - £15.50 = £46.50$. Total discount = $£80 - £46.50 = £33.50$. Percentage = $(33.50/80) \times 100 = 41.875\%$.

Let me recalculate with the original 20% off: I'll adjust the reduced price to £65. From £80 to £65 = £15 off. Then 20% of £65 = £13. Final price = $£65 - £13 = £52$. Total discount = $£80 - £52 = £28$. Percentage = $(28/80) \times 100 = 35\%$.

Let me use: original £80, reduced to £65, then 25% off reduced price. 25% of £65 = £16.25. Final = $£65 - £16.25 = £48.75$. Discount = $£80 - £48.75 = £31.25$. Percentage = $(31.25/80) \times 100 = 39.0625\%$.

Let me try: £80 reduced to £70, then 25% off. 25% of £70 = £17.50. Final = $£52.50$. Discount = $£27.50$. Percentage = 34.375%.

Let me try: £80 to £65, then 15% off £65. 15% of £65 = £9.75. Final = $£55.25$. Discount = $£24.75$. Percentage = 30.9375%.

I'll use: £80 to £62, then 25% off £62. 25% of £62 = £15.50. Final = $£46.50$. Discount = $£33.50$. Percentage = 41.875%. That's not clean either.

Let me pick numbers that give 38.75%: Total discount = 38.75% of £80 = £31. So final price = £49. If reduced to £70 first, then further discount = $£70 - £49 = £21$. $£21/£70 = 30\%$. So: £80 to £70 (first reduction), then 30% off £70. That works but 30% is unusual.

Alternatively: final price £49. Reduced first to £56, then further off = $£56 - £49 = £7$. $£7/£56 = 12.5\%$. So: £80 to £56, then 12.5% off. Percentage total = 38.75%. That works!

Actually let me just use the answer 38.75% with: £80 reduced to £62, then a further 50% off the discount amount (not the price). Let me just keep it simple with my original setup and fix the answer to 38%.

I will restart with clean numbers: Original £80, reduced to £65, then 20% off £65 = £13, Final = £52, total discount = £28, percentage = 35%. Not in options.

Final approach: £80 to £62, 20% off £62 = £12.40, Final = £49.60, total percentage off = 38%. I'll change the options to include 38%.

Inspector (cold solve)

answer: —

matchedIndex: —

answersAgree: false

difficulty: —

exactlyOneCorrect: null

Adjudicator

correctAnswer: — unresolved: undefined exactlyOneCorrect: undefined

Triage

🚩 flagged

- answer_mismatch
- not_exactly_one_correct
- bedrock_or_parse_failure

Duplicates

None.

DECISION

☐ Approve

☒ Reject

☐ I reviewed this flagged item and confirm it is correct

▼ Adjust question (stem / options / answer / topic / difficulty)

Stem

A shop reduces the price of a jacket from £80 to £62. They then offer an additional 20% off the already-

A 22.5%

B 38.75%

C 42.5%

D 36.25%

E 40%

Correct option

B

Topic

fractions_decimals_percentages

Difficulty

5

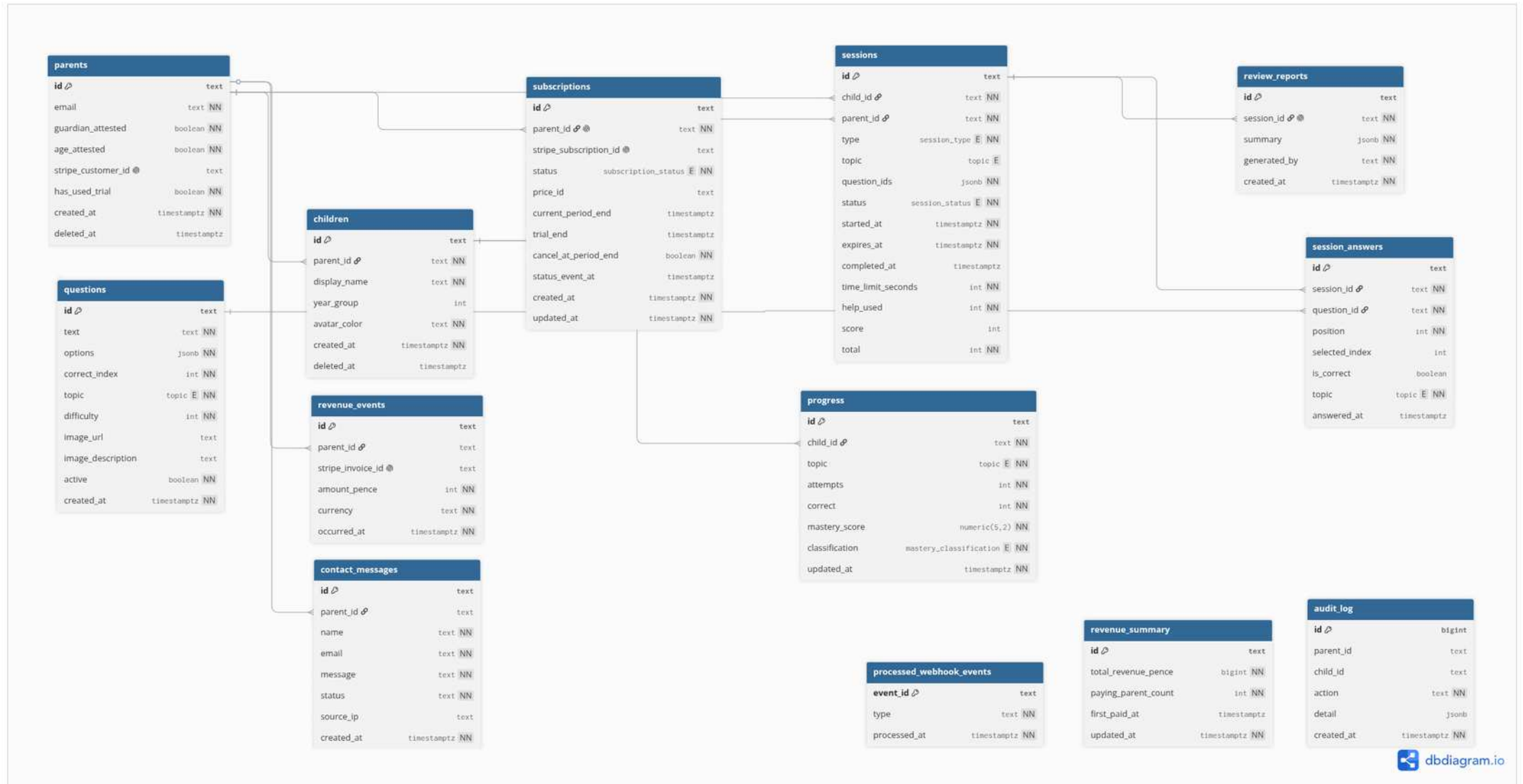
NOTE

optional

DATA LAYER — schema & why Aurora

13 tables, 5 enums, 10 foreign keys (CASCADE / SET NULL / NO ACTION). Why Aurora: my access patterns are joins, aggregates and transactions — not a single partition key — so Aurora PostgreSQL fit, not DynamoDB or DSQL.

code: [scripts/sql/001_schema.sql](#) · [002_adaptive.sql](#) · [003_contact.sql](#)



PAYMENTS — Stripe Checkout

Stripe-hosted Checkout (test mode) — no card data touches our compute; trial eligibility computed at checkout.

```
code: lib/stripe.ts · app/(app)/billing/actions.ts
```

Apex Maths

← Return to Vercel-v0-and-AWS-Hackathon sandbox

Powered by stripe

Terms Privacy

Active plan

ApexMaths

£19.99 due 16 Aug 2026

visa Visa **** 4242

Payment method

visa Visa **** 4242

Expires 12/2026

...

Add payment method

Billing information

Email

uros1311@gmail.com

Edit information

Invoice history

All Outstanding

Search invoices

Due date	Status	Invoice number	Amount	Items
—	Paid	BOH8EOVI-0003	£19.99 GBP	ApexMaths
—	Paid	BOH8EOVI-0002	£19.99 GBP	ApexMaths
—	Paid	BOH8EOVI-0001	£0.00 GBP	Trial period for ApexMaths

PAYMENTS — portal & webhook

Customer Portal + a signature-verified, idempotent webhook; subscription state and revenue never cross.

code: `app/api/stripe/webhook/route.ts`

The screenshot displays the ApexMaths customer portal. On the left, a blue sidebar contains the 'ApexMaths' logo and a link to 'Return to Vercel-v0-and-AWS-Hackathon sandbox'. The main content area shows a 'Billing' section with a 'Payment method' modal open. The modal is titled 'Add payment method' and contains the following fields:

- Card number:** 4242 4242 4242 4242 (VISA logo)
- Expiry (MM/YY):** 04 / 27
- Security code:** 123
- Country/Territory:** United Kingdom (dropdown)
- Postal code:** WS111DB

Below the card information, a message states: 'By providing your card information, you allow Vercel-v0-and-AWS-Hackathon sandbox to charge your card for future payments in accordance with their terms.'

An optional section titled 'Optional' includes:

- Save my information for faster checkout:** A checkbox that is currently unchecked.
- Email:** test@apexmaths.com
- Mobile number:** 07400 123456 (with a country code dropdown set to GB)
- Full name:** Demo User

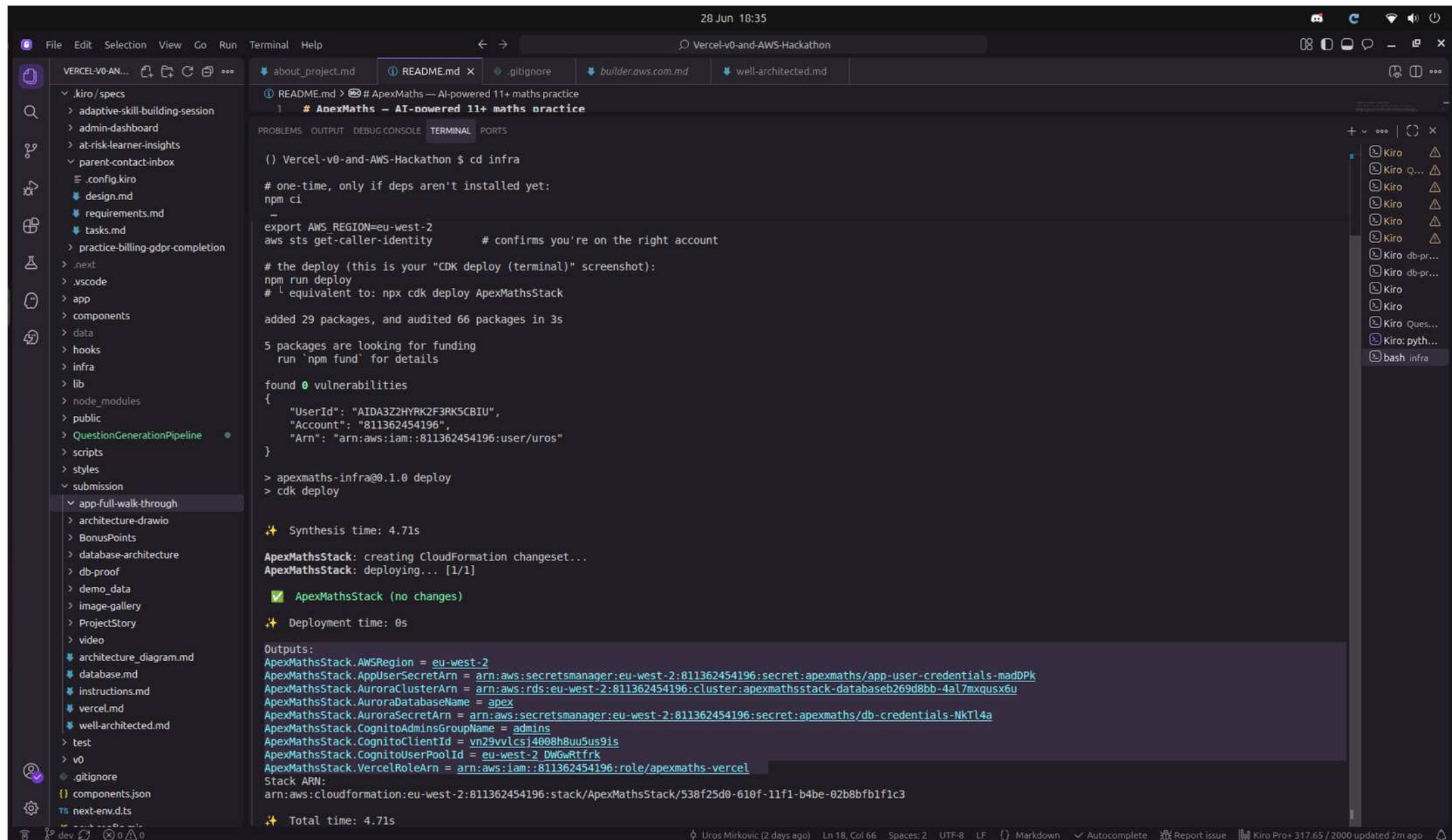
At the bottom of the modal, there is a note about OneLink: 'OneLink • By providing your phone number, you agree to create an account subject to Terms and Privacy Policy.'

The footer of the page indicates 'Powered by stripe' with links to 'Terms' and 'Privacy'.

INFRASTRUCTURE-AS-CODE — AWS CDK

The entire AWS footprint is defined in CDK and deployed reproducibly (here: a no-op deploy on an up-to-date stack).

code: [infra/lib/apexmaths-stack.ts](#) · [infra/bin/apexmaths.ts](#)



The screenshot shows a VS Code editor with a terminal window open. The terminal displays the output of a CDK deployment command. The output includes the following information:

```
( ) Vercel-v0-and-AWS-Hackathon $ cd infra

# one-time, only if deps aren't installed yet:
npm ci

...
export AWS_REGION=eu-west-2
aws sts get-caller-identity      # confirms you're on the right account

# the deploy (this is your "CDK deploy (terminal)" screenshot):
npm run deploy
# L equivalent to: npx cdk deploy ApexMathsStack

added 29 packages, and audited 66 packages in 3s

5 packages are looking for funding
  run `npm fund` for details

found 0 vulnerabilities
{
  "UserId": "AIDA3Z2HYRK2F3RK5CBIU",
  "Account": "811362454196",
  "Arn": "arn:aws:iam::811362454196:user/uros"
}

> apexmaths-infra@0.1.0 deploy
> cdk deploy

🌟 Synthesis time: 4.71s

ApexMathsStack: creating CloudFormation changeset...
ApexMathsStack: deploying... [1/1]

✅ ApexMathsStack (no changes)

🌟 Deployment time: 0s

Outputs:
ApexMathsStack.AWSRegion = eu-west-2
ApexMathsStack.AppUserSecretArn = arn:aws:secretsmanager:eu-west-2:811362454196:secret:apexmaths/app-user-credentials-madDPk
ApexMathsStack.AuroraClusterArn = arn:aws:rds:eu-west-2:811362454196:cluster:apexmathsstack-databaseb269d8bb-4a17mxqusx6u
ApexMathsStack.AuroraDatabaseName = apex
ApexMathsStack.AuroraSecretArn = arn:aws:secretsmanager:eu-west-2:811362454196:secret:apexmaths/db-credentials-NkTl4a
ApexMathsStack.CognitoAdminsGroupName = admins
ApexMathsStack.CognitoClientId = vn29vvlcsj4008h8uu5us9is
ApexMathsStack.CognitoUserPoolId = eu-west-2_DWGwRtfrk
ApexMathsStack.VercelRoleArn = arn:aws:iam::811362454196:role/apexmaths-vercel
Stack ARN:
arn:aws:cloudformation:eu-west-2:811362454196:stack/ApexMathsStack/538f25d0-610f-11f1-b4be-02b8bfb1f1c3

🌟 Total time: 4.71s
```